

Project Statistics & PRH Confirmation Summary

Complete accounting: what the project confirmed, falsified, and bounded — with the fractal thread correctly placed across all three verdict categories. Every number verified against the Final Project Report (84 pp).

1 · Verdicts across all pre-registered predictions

verdict	count	includes
CONFIRMED	21	learned geometry (Exp A/B), cross-modal/lineage transfer (C.8, G4, E.2), hub 93–96% (C.13), encoder ladder (C.14), fractal geometry above random (P1 8/8 both arch, P2 4/4 both arch, P5w 4/4 both arch)
FALSIFIED	7	fractal geometry recoverable through hub — it was not (G10 H1); naive scaling beyond 304M — both ladders fall (E.16 P2/P3); MLP > ridge — gain +0.0000 (F1/F2); P5-cnn raw 0/4
INCONCLUSIVE	4	image-vs-text reality-dependence (P3): CIs straddle zero CNN [−0.106, +0.100], ViT [−0.053, +0.135], raw and whitened
NOT MEANINGFUL	2	H3 CNN/ViT %-of-native ratios blocked by the 0.050 floor guard; both transfer and native at floor — ratio of two near-zero numbers

2 · The fractal thread belongs in three categories, not one

G9 and G10 are a boundary-locating result — not a simple pass/fail. Placing them only in the falsification ledger misrepresents what happened.

Confirmed (G9 — visible geometry)

finding	evidence	verdict
Fractal training creates real shared geometry	0.09 (CNN), 0.21 (ViT) above random	P1 CONFIRMED 8/8 both arch
Fractal > random twin	matched-norm control: 0.067 vs 0.036	P2 CONFIRMED 4/4 both arch
Whitening rescues the collapsed CNN	P5w CONFIRMED 4/4, R _{white} ≈ 11%	P5w CONFIRMED
Fractal signal survives PC1 removal	corrected drop1: 0.060 vs 0.051	ordering holds
Preprocessing confound is minor	norm ≈ 10% of signal, training ≈ 90%	P2-cnn survives corrected

Inconclusive (G9)

finding	evidence
Image-vs-text reality-dependence asymmetry	P3 CIs straddle zero: CNN [−0.106, +0.100], ViT [−0.053, +0.135]. Earlier 'confirmed' withdrawn.

Falsified (G10 — transfer/recoverability)

finding	evidence
Visible fractal geometry is recoverable through the hub as a semantic function	H1 FALSIFIED: transfer at floor (0.014/0.020 ≈ random 0.012/0.017). Natural-image twins transfer at 0.20–0.26.

The fractal probe confirmed the data-driven mechanism (fractals install some structure, proving the 'same reality' clause is load-bearing) while falsifying the strongest reading (that any structured training produces transferable semantic alignment).

3 · What the project confirmed about PRH

PRH clause	what this project showed	strength
"well-trained"	trained > random by 6.3× (CKA); trained curve rises with depth while random decays	strong (Exp A)
"same representations"	80% shared variance, R^2 0.592, cosine 0.900 across labs with zero shared weights/data/tokenizer	strong (Exp A)
"up to a linear transformation"	MLP = ridge exactly ($0.9183 = 0.9183$, gain +0.0000) — exactly linear, not approximately	strong (F1/F2)
"independently"	SigLIP 2 (independent lineage) 94.2%, ConvNeXt (CNN, held out) 96.7% — not a family trait	strong (G4, E.2)
across modalities	image-only → text-only $R@1$ 0.358 = 40.9% of jointly-trained ceiling, 358× chance	moderate (C.8)
scales with capacity	+ 15.3 pts/decade across DINOv2 s/b/l, six monotone measures, R^2 0.949	moderate (C.10)
isotropy is the mechanism	whitening rescues GPT-2 to 102.6% of a contrastive encoder; ladder rank-corr +1.00	strong (C.11, C.14)
self-identifying correspondence	39.7% exact match with zero pairs = 595× chance	moderate (C.12)
one shared space serves all	7 encoders, 42/42 pairs at 103.6%, frozen head transfers to unseen encoders	strong (C.13)
natural-world data matters	fractal geometry exists ($R \approx 10\text{--}21\%$) but doesn't transfer semantically; natural images produce 10–20× more and DO transfer	moderate-strong (G9 + G10)

4 · What the project established that PRH did NOT claim (new)

finding	evidence
Convergence is LOCAL, not global	CKNNA: 21/21 pairs higher locally; CCA: 6 of 64 shared directions; hub R^2 image 0.46–0.57, text 0.02–0.05
Visible ≠ recoverable	ConvNeXt: lowest raw p, highest transfer. Fractal: real geometry, zero transfer. Both directions documented.
No single 'good geometry'	whitening helps reading (+102.6%), destroys finding (39.7% → ~0%) — same transform, opposite effect
The width cliff is numerical, not representational	step at source ambient dim, σ 0.005 vs 10.79, repaired by truncation or regularisation

5 · What the project did NOT establish

claim	status
Universal representation	NOT shown — convergence is local, not global
Image-vs-language asymmetry in reality-dependence	INCONCLUSIVE (P3 CIs straddle zero, both architectures)
Scaling beyond 304M improves alignment	FALSIFIED at this scale (E.16 — both ladders fall)
Fractal training produces semantic transfer	FALSIFIED (G10 — transfer at floor, = random twin)
Global metric identity	NOT shown — shared structure is a small subspace

6 · The honest headline

Confirmed narrowly: trained independent models share mainly local structure that linear maps make useful across modalities and lineages. The relationship is exactly linear, isotropy is the mechanism, and one hub serves all.

Bounded: not global or universal; scale is not monotonic; natural-world data matters; and visible geometry need not transfer.

Project scale

dimension	count
Total encoders touched	14 (8 main roster + MobileCLIP, SigLIP, Pythia, DINOv2-giant, FractalDB ResNet-50, FractalDB DeiT-tiny)
Named experiments	21
Series	10 (A, B, C, E, F, G, G8–G10, H, V, K)
Colab notebooks	~35
COCO items	9,533
Findings	9 headline + 4 implications
Self-corrections	5 documented
Falsified explanations	11 (2 the project's own)
Shipped artifacts	2 (0.79 MB Core ML adapter + Qdrant index)

PDF deliverables	~15
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